CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in AI agents and assistants, business process automation and integrating AI with APIs and data sources, with cloud deployment on Microsoft Azure and AWS. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by three roles at Covestro in the chemical industry working closely with chemists and business teams.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer of an AI agent system for the German energy sector, built end to end (data ingestion, chunking, prompt engineering, indexing), working with SAP domain experts and reporting directly to the CTO.
- Built multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents in parallel) and tested it against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Developed Expert Interface, an internal web app (React frontend, Python microservices backend) on Microsoft Azure where domain experts review and approve knowledge-graph changes.
- Built a Model Context Protocol (MCP) service that connects the knowledge graph as a data source with millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- **Tech stack: Python, Microsoft Azure, LLM APIs (LiteLLM), LangGraph, LangChain, MCP, RAG, GraphRAG, FastAPI, React, Neo4j, MongoDB, Postgres, Docker, Kubernetes**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Advised business stakeholders on how agentic AI could support their decision-making workflows and translated their requirements into a solution design.
- Built and deployed the agentic AI application with LangGraph, routing requests across vector database queries, general inference, and summarization, with a FastAPI backend and React frontend.
- Containerized with Docker, deployed to AWS with CDK, and version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, REST APIs, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated quality testing of an AI assistant over internal chemical data (RAG) with RAGAS and PytrecEval against ground truth written by chemists.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical documents.
- Built the evaluation infrastructure on AWS (Lambda, SQS, DynamoDB) with CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, Azure AI Search, AWS (Lambda, SQS, DynamoDB, S3, CDK)**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data.
- Built a Streamlit app letting chemists simulate parameter changes and see predicted outcomes before running physical batches.

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.

PROJECTS

Project: Quality Process Copilot | Copilot Studio | Power Platform | Azure AI Foundry

- Copilot in Microsoft Copilot Studio that answers employees' questions from pharma SOP documents through an Azure AI Foundry model with retrieval, and starts a deviation report when something goes wrong in production.
- Process automation on the Microsoft Power Platform: a Power Automate flow routes each deviation report for review and approval and stores it in SharePoint; a Power Apps screen gives quality managers the overview, built as reusable components with documented standards and best practices.

TECHNICAL SKILLSET

- **AI Agents & Copilots:** AI agents and multi-agent systems (LangGraph, LangChain), LLM APIs (LiteLLM), prompt engineering, RAG, GraphRAG, MCP, evaluation (RAGAS, PytrecEval), Claude Code
- **Microsoft & Cloud:** Microsoft Azure (web app deployment, Azure AI Search), AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3), Docker, Kubernetes, GitLab CI/CD
- **Integration & Development:** Python, REST APIs (FastAPI), Postman, React (beginner), Streamlit, SQL
- **Data Sources:** Postgres, MongoDB, Neo4j, DynamoDB, vector databases

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)